

By the way...
during the First World War, I worked on ways of detecting submarines using sound waves.

Ernest Rutherford

The man who discovered the structure of the ATOM, split it apart, and won a Nobel Prize in Chemistry

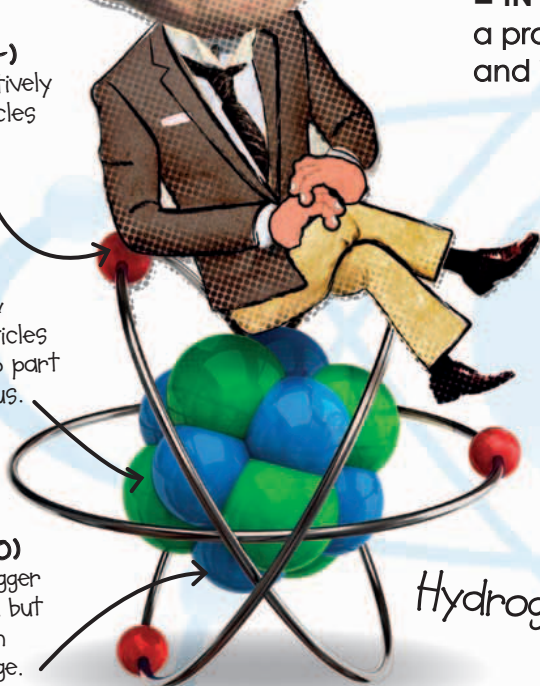
All about me

- **BORN:** 1871
- **DIED:** 1937
- **NATIONALITY:** New Zealander
- **FACTOID:** I am sometimes called "the father of nuclear physics."
- **IN A NUTSHELL:** Born in New Zealand, I became a professor of physics at McGill University in Canada and investigated the new discovery of radioactivity.

Electrons (-)
are tiny negatively charged particles that orbit the nucleus.

Protons (+)
are positively charged particles that make up part of the nucleus.

Neutrons (0)
are slightly bigger than protons, but don't have an electric charge.



Particle puzzle

After Marie Curie proved that **atoms** weren't just solid balls, Rutherford figured out that most of an atom's mass is at its center, or **NUCLEUS**, and the rest of the atom is actually made up mostly of empty space. He also realized that the nucleus is made up of smaller particles, called **protons** and **neutrons**, surrounded by a cloud of tiny particles called **electrons**.

Hydrogen is the only element without any neutrons

Splitting the atom

Rutherford discovered that an atom's nucleus could be **smashed apart** if it is struck by another high-energy particle (like billiard balls smashing together). Using a radioactive source to fire particles at an atom, he found that impact knocked the protons out of the nucleus and **split** the atom. Rutherford had created a new science—**NUCLEAR PHYSICS**.

